BIRZEIT UNIVERSITY

Faculty of Engineering & Technology Electrical & Computer Engineering Department

Interfacing techniques – ENCS4380

Homework 1

E-nose

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I. Intelligent system sensors (ISS)

An Intelligent Sensor System (ISS) is a device that goes beyond simple sensing by integrating its core components: a **sensor** that responds to a stimulus, a **system** of cooperating processes, and **intelligence**, which is the ability to combine prior knowledge with adaptive learning

If we combine the sensing element with a signal processor this is when we have an intelligent sensor, reprocessing the signals will need compensating for the temperature and power supply , to balance and also filtering is needed of the signals and the non-linearity that we don't want. [3]

E-nose

Is a modular system device that is capable of sniffing and determining odours components, It is also seen as an array of sensors that generate signals to the volatile organic compounds VOC in the odour examined. [1]

How does it work

Sample delivery system : The volatile organic components in the odour here is being sniffed and transferred through a pipe to the array of sensors. **Detection system** : Here reactions on each slot happen based on the components, For example if the odor is composed of X,Y and Z we have at least 3 slots for each component that won't react unless the odor has the exact same component. The adsorption of VOCs on the sensor surface causes a physical change, and this is transduced into electrical signals (a change in electrical properties/resistance/frequency). **Computing system** :After the components are detected and corresponding to us humans when we have the nerve signals transferred to the brain we have the ANN using AI to analyze these reactions and determine the odour components. It compares the odour with the odours that we have in databases and recognizes patterns where each odour has its ID.[1]

II. Case Study : Food allergen/Lard detection

detection of Volatile Organic Compound (VOC) associated with (lard) and alcohol in processed food

People with allergies and people of religious background and even vegetarians of all kinds struggle when they are travelling or staying away and don't know what to eat. This was my own experience when I went to Italy as an exchange student and I struggled with determining if the food I'm eating had any non'halal ingredients specially since many pastries in Italy like Cannolis most of the time contain lard and I even struggled communicating to the chefs/restaurant owners. It doesn't stop at lard. Many bread products also have a significant amount of alcohol that many of us don't want to consume.

This problem is relevant because relying on human assurances is not trustable and sometimes even impossible. Restaurant staff and cooks usually do not know the full ingredient list, may falsely claim compliance to ensure a sale, and printed labeling is often untrustworthy.When you have human error you need the machine to solve it!

Our goal with this device is to provide an accurate testing to detect the targeted food (e.g., pastries, breads, pre-mixed ingredients) if there's lard known as *porcine-derived fa*t and alcohol known as *ethanol* present. The system must deliver an objective result right away, informing consumers, restaurant staff, or compliance officers whether a sample meets strict dietary standards.

Target VOCs (The Chemical Fingerprint)

To successfully detect alcohol and lard we need a smell print that is created by the VOCs that are released by alcohol and lard .

Target Contaminant	Key Volatile Organic Compounds (VOCs) for Detection
Lard (Porcine Fat)	Aldehydes & Ketones: Lard, especially when heated or slightly aged, releases aldehydes such as Hexanal, Heptanal, and Nonanal. These are products of lipid oxidation, and their specific concentration profile forms the unique "lard signature" for the E-Nose to learn. The presence of ketones like 2-Pentanone and 2-Hexanone also contribute to the unique signature.[4]

Alcohol (Ethanol)	Ethanol and Byproducts: In yeast-leavened products (like bread and some pastries), the primary volatile compound released during baking is Ethanol (C_2H_5OH). The E-Nose can be specifically trained to detect ethanol itself, as well as fermentation byproducts like Acetoin (3-Hydroxybutan-2-one), which is strongly associated with yeast activity and spoilage.[5]

Sampling Methodology

Our E-Nose needs a clear and controlled way to "sniff" the food and catch a strong scent from the lard and alcohol. To do that we first prepare the sample by heating it. Since fats and alcohol usually hide their smell when they're cold, the sample has to be gently warmed so the scent can come out clearly. We'd take a small piece of the food like a bit of Cannoli or a crust of bread and place it in a small, disposable container with a built-in heater. This heater quickly raises the temperature to a medium level (around 60–80 °C) which helps release the volatile organic compounds (VOCs) without burning the food. Heating the sample like this makes the VOCs come into the air, creating a centralized scent that's easy for the sensors to detect [1].

Once the air above the sample is filled with VOCs, a small pump inside the E-Nose pulls in that air and directs it over the sensor array, and then the device captures the unique scent "fingerprint" of the food [1].

After the reading is taken, the system quickly moves to the cleanup phase. A flow of clean air called the purge is passed through the sample chambers to clear out any leftover odor molecules. This makes sure the smell from one pastry doesn't interfere with the next one and keeps the sensors ready for the next test. [1].

System Challenges

The main challenge for this E-Nose is called the **food matrix effect**. Foods like Cannolis for example often contain thousands of VOCs. The ones that we aren't interested in checking will act like noise making it harder for us to identify lard and alcohol. to work correctly we must prove that the fat we have for example is lard and not just any fat.

To overcome these challenges we must have the specialized hardware and the advanced software in our ISS.

1. Sensor Array

Our device uses a non-selective array of chemical sensors, such as Metal Oxide Sensors (MOS) or Conducting Polymers.

The key is that each sensor responds in a unique way to many different chemicals, which, when combined, produces a complex but unique odor fingerprint for the *entire* food matrix (including the noise).[1]

2. Training and Adaptation (A Priori Knowledge)

The system's "intelligence" comes from an Artificial Neural Network (ANN). This ANN must go through intensive training in the laboratory aka A Priori Knowledge stage.

During training, the ANN learns to recognize and filter out the constant "noise" patterns (e.g., the smell of sugar and flour) and learns to associate the remaining, subtle signal specifically with the signature of the target contaminants (lard and ethanol).[1]

3. Decision Making: Pattern Recognition

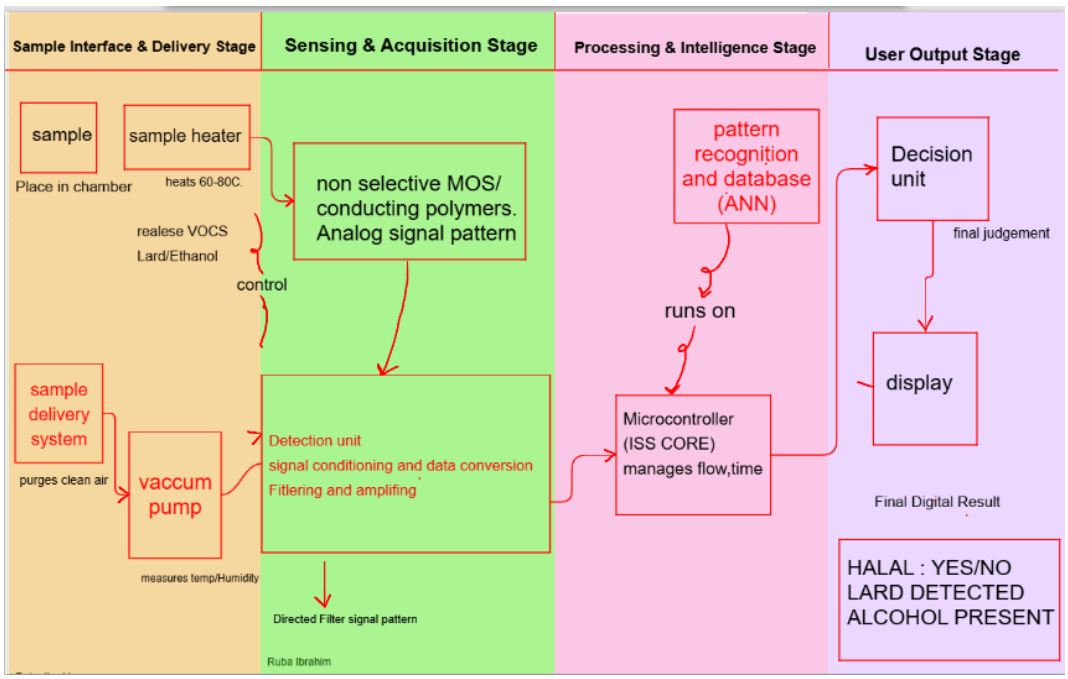
In the field, when a user tests a food item, the ANN performs complex pattern recognition on the complete response pattern from the sensor array.

It automatically corrects for environmental factors (like temperature or humidity compensation) and compares the corrected signature against its Halal/Non-Halal/Alcohol database.

The final output is simple and actionable, providing an instantaneous result like: "Contaminant Present: YES" or "Halal Compliant: YES/NO".

III. Block diagram

The process begins when a food sample is heated, releasing Volatile Organic Compounds (VOCs) that are then drawn by a vacuum pump across a non-selective sensor array. This array generates a unique analog "fingerprint" which is immediately amplified, filtered, and converted into a clean digital signal. This digital data is then processed by the microcontroller, which runs an Artificial Neural Network (ANN) to compare the signal against a pre-trained database while compensating for environmental factors. Finally, a decision unit applies a threshold to the ANN's analysis and generates a simple, actionable result for the user on the display.



IV. Critical Reflection

While the electronic nose (E-Nose) is a powerful example of an intelligent sensor system, its application is filled with challenges. For it to be reliable it must provide us with accurate results under different conditions, there are 2 limitations that will be discussed here: sensor **drift** and **interference from environmental Factors**,. the "unknown gas scenario" will be considered as well.

Limitation 1: Sensor Drift and Long-Term Stability

A very well-known limitation of E-Nose systems is **sensor drift**, which affects the accuracy of the

algorithms overtime. It involves deviation of its VOC-binding characteristics over its operational lifetime.

Scientific Cause : Research highlights that even short-term sensor drift can have a crucial effect on results, caused by subtle daily variations in temperature, humidity, and gas flow. These factors alter how VOCs interact with the sensor array, meaning an odor's unique "fingerprint" can change from one day to the next, producing incorrect patterns even with standardized procedures.[6]

Impact on the Unknown Gas Scenario : Sensor drift makes identifying unknown substances exceptionally difficult because it alters the signal patterns of known gases, making them inconsistent with the system's pre-trained database. A clinical study showed that this drift can be a stronger signal than the actual disease biomarkers being measured. This causes the E-Nose to misinterpret a familiar but drifted signal as a new, unknown substance, undermining its core reliability.[6]

Suggested Improvements :The study emphasizes that correction for sensor drift is not yet routinely performed in clinical studies but is essential for reliability and repeatability. Methods include:

1. **Drift Compensation Algorithms**:Software algorithms are essential for correcting sensor drift. The IBD study improved accuracy to 0.68 using logistic regression, while other methods like PCA and machine learning create a more resilient "virtual sensor"[6].
2. **Standardized Measurement Protocols with Baseline Correction**: A process-based solution is to improve the measurement protocol by regularly testing 'blank' samples of filtered air. This allows the system to track the sensor baseline and apply correction factors, effectively recalibrating itself during operation.[6].

Limitation 2: Interference from Environmental Factors

E-Nose sensors are highly sensitive not just to target VOCs, but also to environmental conditions, particularly humidity and temperature. These environmental issues act as interferences that can change a sensor's response pattern, even with constant gas composition, leading to inaccurate readings and

potentially invalidating a trained discriminant model [7].

Scientific Cause : Sensors, particularly MOS types, are very sensitive to their surroundings. Changes in temperature and humidity alter their readings. For example, water vapor in the air can block the target smell from reaching the sensor, and temperature changes the sensor's reaction speed. This means the same smell will produce a completely different "fingerprint" in a hot, humid kitchen than it would in a cool, dry lab.[7]

Impact on the Unknown Gas Scenario: Environmental interference makes it almost impossible to identify unknown gases. An E-Nose trained in one environment will be unreliable in another. If it detects an unfamiliar signal in a humid room, it can't know if it's a genuinely new substance or just a familiar smell that has been distorted by the humidity. This ambiguity leads to incorrect results, making the device untrustworthy for discovering new contaminants.

Suggested Improvements : The research paper by Liu et al. categorizes interference suppression techniques into several types. Based on their review, two effective strategies for countering environmental factors are interference compensation models and hardware optimization.

1. Interference Compensation Models: One solution is to build a software model to cancel out environmental effects . This is done by adding temperature and humidity sensors to the E-Nose . Data from all sensors is then fed into an AI model (ANN). By training the AI in many different conditions, it learns how heat and humidity distort the smell and can mathematically remove these distortions from the signal, making the E-Nose more accurate [7].
2. Hardware Optimization:Another solution is to use hardware to create a stable environment for the sensor, making it independent of outside conditions. A simple way is to use a small heater to keep the sensor at a steady temperature . A more advanced method, called temperature modulation, involves rapidly cycling the sensor's temperature during a measurement. This creates a much more detailed 3D "fingerprint" of the smell, which is less affected by environmental interference.[7]

Personal Insight and conclucion

As a student to see how these systems can be designed and implemented to offer practical solutions that enhance public health, safety, and cultural practices makes me excited and proud each time I remember why I chose Computer Engineering,

One thing where we can apply this system is in public health. As someone who suffers from sensitive sinuses and asthma, where air quality is a daily concern. A portable and affordable E-Nose device could really help me and many like me to detect how polluted the air testing the VOCS and this device can be paired with phones to send a warning to wear masks.

Furthermore, I have explored more into food safety in my study case and tried my best to search for the relevance with my community. The challenge of identifying lard in food for religious reasons is a daily reality for many of us who travel/have travelled. My recommendation was a device that detects the VOC fingerprints of common allergens, such as peanuts or gluten, which are often hidden in prepared foods ,lard and alcohol.

conclucion

This project provided me with an excellent understanding of E-nose as a complete system, it put me in the experience of integrating researches and depending fully on me. I went through a comprehensive understanding of the electronic nose, not just as a sensor, but as a complete Intelligent Sensor System.

A critical takeaway is the importance of the **a priori knowledge** present in the system's Artificial Neural Network (ANN). As in the case study, the E-Nose's ability to reliably detect components like lard or alcohol is entirely dependent on the quality and breadth of its training data. The ANN must be taught to isolate the specific VOC fingerprint of a target substance from a complex and noisy background.

Finally, this homework has shown the gap between a lab prototype and a real-world device, where critical challenges like **sensor drift** and **environmental interference** must be overcome with advanced

compensation mechanisms

Resources

Note : Gemini AI the normal version was used to paraphrase some paragraphs, to tell if I missed anything and to evaluate my work based on the criteria mentioned in the homeworks original document.

1. Course Slides. *Interfacing Techniques*. Provided by instructor.

2. Taylor & Francis. *Intelligent Sensor*. Taylor & Francis Knowledge Base. Accessed: 2025-10-17. https://taylorandfrancis.com/knowledge/Engineering_and_technology/Industrial_engineering_%26_manufacturing/Intelligent_sensor/

3. Yue, M. (2019). *Intelligent sensors: Structure and system*. ResearchGate. https://www.researchgate.net/publication/337965825_Intelligent_sensors_Structure_and_system

4. Zhai, E., Zhang, J., et al. (2025). *Formation Mechanism of Lipid and Flavor of Lard Under the Intervention of Heating Temperature*. Foods, 14(14), 2441. https://www.mdpi.com/2304-8158/14/14/2441

5. U.S. Environmental Protection Agency (EPA). *Site 4—Determination of VOC, Ethanol, and Acetaldehyde Emissions from Commercial Bakeries*. https://gaftp.epa.gov/ap42/ch09/s096/related/emission_test_report_42tr012s.pdf

6. Bosch, S., et al. (2022). *Influence of Short-Term Electronic Nose Sensor Drift on Fecal Volatile Organic Compound Profiles in a Real-Life Inflammatory Bowel Disease Cohort*. Sensors, 22(23), 9246. https://pmc.ncbi.nlm.nih.gov/articles/PMC9740993/

7. Liu, T., et al. (2018). *A Review on Interference Suppression in E-Nose for Gas Sensing*. Sensors, 18(4), 1179. https://www.mdpi.com/1424-8220/18/4/1179